



Variable Acceleration · Worksheet 1

Mechanics Chapter 4: Variable Acceleration. Answer all questions. Show your working clearly. Total: 63 marks. Time: 80 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 4. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- In every question a particle P moves in a straight line; s (or x) is in metres, v in m s^{-1} and t in seconds, with $t \geq 0$.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Differentiating to find velocity and acceleration	17	
B Integrating to find velocity and displacement	20	
C Stationary values and distance travelled	21	
Roar-level challenge	5	
Total	63	

Section A Differentiating to find velocity and acceleration

17 marks

1. The displacement of P is $s = 4t^2 - 3t$. Find the velocity of P when $t = 2$. [2]

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2. The velocity of P is $v = t^3 - 2t$. Find the acceleration of P when $t = 1$. [2]

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3. The displacement of P is $s = 2t^3 - 9t^2 + 12t$. Find the times at which P is instantaneously at rest. [3]

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4. The displacement of P is $s = t^3 - 6t^2 + 5$. [3]

- (a) Find the acceleration of P when $t = 3$.
- (b) Find the speed of P at the instant when its acceleration is zero.

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5. A particle P moves along the x -axis. At time t seconds, its displacement from O is x metres, where $x = t^3 - 12t + 3$.

(a) Find an expression for the velocity of P .

[1]

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(b) Find the time at which P is instantaneously at rest, and its displacement from O at this time.

[3]

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(c) Find the acceleration of P at this time.

[1]

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(d) Find the speed of P when $t = 3$.

[1]

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(e) Describe the motion of P just before and just after the time found in part (b).

[1]

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(Total for Question 5 is 7 marks)

**Section B Integrating to find velocity and displacement**

20 marks

6. The acceleration of P is $a = 6t$. When $t = 0$, $v = 2$. Find an expression for v . [2]

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7. The velocity of P is $v = 4t - t^2$. When $t = 0$, P is at O . [3]
Find an expression for the displacement of P from O , and the displacement when $t = 3$.

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8. The acceleration of P is $a = 12 - 6t$. P starts from rest. [3]
Find an expression for v , and the velocity when $t = 1$.

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9. The acceleration of P is $a = 2t - 4$. When $t = 0$, $v = 3$ and P is at O . [4]
Find expressions for v and s , and show that P returns to O when $t = 3$.

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10. A particle P starts from O . At time t seconds, its acceleration is $(6t - 4) \text{ m s}^{-2}$. When $t = 0$, the velocity of P is -2 m s^{-1} .

(a) Find an expression for the velocity of P . [2]

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(b) Find an expression for the displacement of P from O . [2]

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(c) Find the displacement of P from O when $t = 3$. [1]

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(d) Find the displacement of P between $t = 1$ and $t = 3$. [2]

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(e) Find the time at which the acceleration of P is zero. [1]

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(Total for Question 10 is 8 marks)



Section C Stationary values and distance travelled

21 marks

11. The velocity of P is $v = 6 - 2t$. Find the time at which P is at rest. [2]

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12. The velocity of P is $v = t^2 - 6t + 8$. Find the minimum velocity of P , and the speed of P at that time. [3]

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13. The velocity of P is $v = 3t^2 - 12t + 9$. Find the distance travelled by P between $t = 0$ and $t = 1$. [3]

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14. Using the velocity in Question 13, find the total distance travelled by P between $t = 0$ and $t = 3$, and its displacement over this interval. [4]

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15. A particle P moves along a straight line. At time t seconds, its velocity is $v \text{ m s}^{-1}$, where $v = 12t - 3t^2$.

(a) Find the maximum velocity of P , justifying that it is a maximum. [3]

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(b) Find the time at which P comes to rest. [1]

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(c) Find the distance travelled by P in the first 4 seconds. [2]

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(d) Find the total distance travelled by P in the first 6 seconds. [3]

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(Total for Question 15 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. A particle moves in a straight line with velocity $v = t^2 - 5t + 4 \text{ m s}^{-1}$ at time t seconds. [5]
Find the total distance travelled by the particle between $t = 0$ and $t = 5$.

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END OF WORKSHEET

Total: 63 marks